CARLIE GONZALEZ

EMBEDDED SYSTEMS ENGINEER

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SUMMARY

Software Engineering student, graduating in May 2027. Hands-on experience developing embedded systems using C++, ESP32, STM32, and Arduino. Skilled in microcontroller programming, hardware integration, and UART/I2C communication. Seeking an embedded software or firmware engineering position.

PROFESSIONAL EXPERIENCE

Treobytes Facilitator- San Diego, CA **July 2025- Present**

- Deliver hands-on coding, robotics, and drone instruction to middle school students, applying programming, electronics, and engineering concepts.
- Configure, operate, and troubleshoot robotics and drone equipment while collaborating with team members to support technical programs across multiple locations.

PROJECTS

ESP-32 MP3 Player| C++, Arduino IDE, UART, I2C

- Developed an ESP32-based MP3 player in C++, integrating a DFPlayer Mini audio module, OLED display, and physical playback controls.
- Implemented UART and I2C communication to control audio playback, update the display, and manage user input through GPIO buttons.

STM32 LED Control System | C++, Nucelo-F401RE GPIO, Interrupts

- Developed a microcontroller-based LED control system using STM32, implementing GPIO inputs and outputs.
- Programmed button-controlled LED functionality using interrupts to manage hardware interactions.

Smart Baking Assistant| React, JavaScript, CSS

- Collaborated on the development of a smart kitchen assistant designed to integrate a digital food scale with an interactive recipe application.
- Developed a React-based recipe interface featuring ingredient instructions, search functionality, and filtering to support future hardware integration.

TECHNICAL SKILLS

Programming Languages: C++, Python, Java
Microcontrollers & Hardware: ESP32, STM32, Arduino, GPIO, Sensors
Communication Protocols: UART, I2C
Development Tools: Arduino IDE, Git, GitHub, VS Code
Concepts: Embedded Systems, Interrupts, PWM, Hardware-Software Integration

EDUCATION

California State University San Marcos **Expected May 2027**
Bachelor of Science, Software Engineering

PROFESSIONAL LINKS

Portfolio- https://carlie-portfolio.vercel.app/
GitHub- https://github.com/carliejlynne
LinkedIn- https://www.linkedin.com/in/carlie-gonzalez/